



Computational Thinking and Artificial Intelligence

Class 7

Student Handbook



First Edition: March, 2026

Country of Publication: India

Published by: Central Board of Secondary Education, Integrated Office, Sector 23, Dwarka, New Delhi-110077

PREFACE

The National Education Policy (NEP) aims to position India as a leader in emerging knowledge fields by integrating technologies like AI, Machine Learning, Big Data, and Computational Thinking into school education. It promotes technology-enabled, interactive, and gamified learning using tools such as Augmented Reality (AR), Virtual Reality (VR), and virtual labs to foster creativity, problem-solving, and interdisciplinary exploration. NCFSE 23 carries this recommendation further for implementation.

While Artificial Intelligence (AI) is an important requirement, Computational Thinking (CT) should be a broader skill, developing a foundation for learning AI. It can cover various aspects like Cybersecurity, basic networking, etc. Hence, CBSE approaches this by integrating Computational Thinking with AI and other technological advancements, without dependence on any platform.

The book engages learners with problems involving multi-layered constraints, conditional dependencies, optimisation strategies, data interpretation and structured decision-making across numerical, spatial and real-life contexts. It deepens Artificial Intelligence understanding through concepts such as rule-based classification, data representation, bias and fairness and decision-making systems, enabling students to analyse how data, rules and assumptions influence intelligent behaviour. The document also provides pedagogical guidance, learning resources, assessment support and classroom implementation guidelines to facilitate competency-based learning in alignment with NEP 2020.

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The efforts of Prof. Manish Jain and his whole team from Centre for Creative Learning, IIT Gandhinagar are also acknowledged and deeply appreciated.

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Introduction

Computational Thinking (CT) is a problem-solving approach that comprises Decomposition, Pattern Recognition, Abstraction, Algorithm Design, Data Analysis and Troubleshooting. Computational Thinking Skills involve solving complex problems that promote thinking skills such as critical & creative thinking, abstraction and pattern recognition, as well as algorithmic thinking. Problem identification and problem solving necessitate the application of multidisciplinary understanding for creating effective solutions.

Artificial intelligence (AI) is a cutting-edge technology that empowers machines and computers to perform tasks that usually require mimicking human intelligence. These machines can perform complex thinking processes such as data analysis, pattern recognition, prediction of trends, solving problems and decision making. Thus, AI involves simulating cognitive processes associated with human intelligence and is widely applicable in various sectors such as banking, healthcare, defence, education, entertainment, agriculture and others for processing information, solving intricate problems and for planning.

The National Education Policy (NEP) aims for India to emerge as a global leader in new emerging knowledge domains such as artificial intelligence, machine learning, data analytics, 3-D machining etc. To realise this goal, the policy suggests teaching students Mathematics and Computational Thinking, along with new subjects like Artificial Intelligence, Machine Learning, and Data Science during their school education. The policy also focuses on technology-enabled learning and classrooms by using tools like artificial intelligence, machine learning, and adaptive testing to create knowledge.

The National Curriculum for School Education draws from this policy aspiration and emphasises the need to introduce these emerging domains of study and technologies in the school curriculum. It recommends inclusion of subjects such as design thinking, augmented reality, virtual reality, artificial intelligence, and computational thinking. Additionally, it promotes the use of gamified content, interactive content, and immersive experiences (such as AR, VR, or virtual labs) to enhance student learning. In a variety of subjects, including design, music, art, and sciences, these resources support students in knowledge creation and exploration, and development of capacities such as problem-solving, critical and creative thinking.

CBSE, under the aegis of the Department of School Education and Literacy, Ministry of Education, Govt. of India, is implementing a Curriculum on Computational Thinking and Artificial Intelligence (CT & AI) to inculcate AI-readiness in school students. This curriculum will be implemented from classes 3rd to 8th, in the session 2026-27, and aims to develop AI-Ready learners, by focussing on Computational Thinking Skills. The AI-readiness, so inculcated through CT Skills, will help develop the capacities of learners to use computational thinking, such as logical thinking, problem solving, pattern recognition, and so on, and understand the role and use of Artificial Intelligence in daily life. The Curriculum aims to build strong foundations in computational thinking, digital literacy, and responsible use of technology, along with nurturing innovation, critical thinking, and ethical decision-making capacities.

1. Relevance: Importance of introducing Computational Thinking (CT) and Artificial Intelligence (AI)

- **Preparing for the future:** To contribute to the world of work in modern societies, individuals need capabilities such as problem solving, using data effectively, identifying patterns, and applying AI ethically for various purposes in life.
- **Holistic Development:** Study of CT and AI contributes to development of reasoning, logical thinking, creative problem-solving skills, critical thinking, ethical decision-making abilities, leading to individual flourishing. It leads to creation of responsible digital citizens in society.
- **Interdisciplinary Relevance:** Embedding CT and AI concepts in the school curriculum helps students to develop an integrated view of the world by connecting various disciplines such as Mathematics, Science, Humanities etc.
- **Innovation and Entrepreneurship:** At the core CT and AI is about solving problems, devising innovative solutions and recreating human thinking. This leads to an entrepreneurial and innovative mindset in the learners.
- **Ethical Awareness:** Study of CT & AI will sensitise learners about the misuse and bias, fairness and inclusivity in AI systems.

2. Objectives (Curricular Goals)

- **CG-1:** Develops skills and capacities of computational thinking, namely, decomposition, pattern recognition, data representation, generalisation, abstraction, and algorithms to solve problems where such techniques of computational thinking are effective.
- **CG-2:** Develops spatial and visual reasoning.
- **CG-3:** Gain foundational knowledge of AI, its types, and domains.
- **CG-4:** Understand key ethical terms such as bias and fairness in relation to AI.
- **CG-5:** Demonstrates proficiency to use Computer & other devices, computer applications for learning and practical purposes such as data analysis, preparation of visual representations and communication of ideas.

3. Learning Outcomes

Computational Thinking (CT) Learning Outcomes

ABSTRACT THINKING

Students will be able to interpret and solve complex, multi-layered problems by:

- Visualising and analysing 3D objects and their transformations, including rotations, reflections, cross-sections, and nets.
- Understanding compound transformations involving multiple flips, turns, folds, and rearrangements.
- Identifying hidden relationships and constraints within incomplete figures, patterns, or logical setups.
- Analysing symmetry, congruence, and proportional reasoning across different representations.
- Interpreting relative positions, orientations, and viewpoints of objects in advanced visual scenarios.

PATTERN RECOGNITION

Students will be able to recognise, extend, and predict complex patterns involving:

- Multi-rule numerical sequences, including alternating, nested, and dependent patterns
- Algebraic patterns using variables, expressions, and functional relationships
- Visual and geometric patterns formed through transformations or growth rules
- Letter and symbol-based patterns involving positional and logical dependencies
- Integrated patterns combining numbers, shapes, symbols, and logical conditions

DECOMPOSITION

Students will be able to break down real-world and abstract problems by:

- Separating interconnected conditions and constraints into manageable components
- Analysing number properties (factors, multiples, ratios, percentages, powers) within layered clues
- Deconstructing problems involving spatial reasoning, measurements, and geometry
- Interpreting tables, grids, charts, and flow-based information with multiple dependencies
- Breaking multi-step logical situations (movement, exchanges, comparisons, scheduling) into ordered steps
- Translating visual or verbal information into structured data for systematic analysis

ALGORITHMIC THINKING

Students will be able to design and follow logical procedures to solve advanced problems involving:

- Rule-based sequences and algorithms with conditional branching
- Grid-based navigation and pathfinding with constraints and decision points
- Step-wise transformations involving calculations, swaps, transfers, or positional changes
- Ordering and arranging elements (people, objects, events) using multiple attributes and logical clues
- Solving problems using if-then reasoning, elimination strategies, and logical consistency checks
- Creating or analysing procedural steps to reach an optimal or valid solution

Artificial Intelligence (AI) Learning Outcomes

Learners will be able to:

- Distinguish key predictive techniques such as:
 - **Regression:** The method of predicting a number based on patterns in past data
 - **Classification:** The process by which a machine arranges things in a group based on what it has learned
 - **Clustering:** The process by which a system automatically puts similar items together
- Explain about the key domains of AI, namely:
 - **Data Science:** learn to manage and extract insights from data
 - **Computer Vision:** learn the basics of how machines understand and respond to visual information
 - **Natural Language Processing:** understand the basics and limitations of how computers process and handle natural language inputs
- Explain what **bias in AI** means, and identify situations where AI can give unfair results
- Demonstrate courteous, safe, and responsible use of technology as part of good digital citizenship

- Use safe practices for maintaining data privacy, including giving informed consent before personal data is collected, used, shared, archived, or deleted
 - Collect and organise simple structured data, interpreting patterns and trends, and create bar charts, line graphs and pie charts
 - Apply basic predictive approaches/techniques to a small dataset
 - Explain uses of AI in healthcare, education, transport, and communication

4. Mapped with NEP and NCF 2023

- The National Education Policy (NEP) aims to position India as a leader in emerging knowledge fields by integrating technologies like AI, Machine Learning, Big Data, and Computational Thinking into school education
- NCFSE 23 carries this recommendation further for implementation
- Learning standards are derived using the approach suggested in and are aligned to NCF SE-2023
- Curricular Goals are derived from the Aims of Education, along with other relevant considerations

5. Time Allocation

The Middle stage (Class 6–8) suggests 100 hours annually, including the time for specific topics on basics of AI and interdisciplinary projects. For Grade 7, the breakdown is:

- **Advanced CT skills:** 40 hrs. per academic year
- **Introductory concepts of AI:** 20 hrs. in an academic year
- **Interdisciplinary projects:** 40 hrs. total (20 hrs. allocated to each of the two required projects)

6. Approach / Pedagogy

- Use hands-on and real-world problems, collaborative and group work to solidify and apply multidisciplinary foundational knowledge on Coding, Data Analysis and Artificial Intelligence tools
- Use complex puzzles, riddles and games to build on the computational thinking abilities taught in the previous stage
- Deliver fundamental concepts of AI through explanations, demonstrations and hands-on experience
- Organise group discussions, design collaborative projects that integrate CT & AI, and offer guidance to students to carry out these projects
- Independent activities for students such as data collection, organisation and analysis created using digital tools or manually
- Discussions, debates and case studies on ethical use of AI

7. Assessment

Assessment approaches move away from traditional summative assessment to continuous, formative, and competency-based assessment. Methods for Class 7 include:

- Written Tests and Practical Examinations
- Project presentations, assignments, and reflective journals

- Interactive Group activities like treasure hunts
- Teacher Observation Journal
- Thematic Projects and Reflections/Group Discussions

CT and AI Transition

- Computational Thinking (CT) forms the basis of learning AI.
- Skills like breaking problems into parts, spotting patterns, filtering essential information, and designing step-wise procedures are the same reasoning processes that power AI and ML systems.
- The curriculum begins with the introduction of CT and deepens it as we move across the stages; AI is introduced later, once pre-requisite knowledge of CT is built for understanding AI.

How to Use This Book?

PART-1 Computational Thinking

The part 1 of this handbook is designed as a companion to the Mathematics textbook and is intended to be used alongside regular classroom teaching. Since it follows the same chapter sequence, the Mathematics teacher can seamlessly integrate it into daily instruction. As concepts are introduced in class, the corresponding questions from this book can be used to deepen understanding and encourage application.

Before beginning a chapter, the teacher is encouraged to read and identify the underlying concepts required for each question and plan how to align them with classroom teaching. As these concepts are taught, the teacher can introduce the related 'thinking questions' to students. It is important to note that the questions in this book are thinking-based and designed to promote analysis, reasoning, and problem-solving.

Teachers should adopt a facilitative approach, guiding students through prompts and discussions rather than directly providing solutions. Students should be given time to think and attempt independently, followed by classroom discussions where different approaches are shared and explored.

Some chapters also include activities that build intuition and engagement. These should be conducted before attempting the questions, as they help students approach the problems with better understanding.

PART-2 Artificial Intelligence

The part-2 of the handbook provides a structured introduction to Artificial Intelligence (AI) as a technology that enables machines to learn from data, recognise patterns, and make decisions. The concepts of AI are presented using simple explanations and real-life examples from areas such as healthcare, education, transport, and communication.

Each chapter includes:

- ▶ Foundational understanding of AI concepts
- ▶ Real-life examples and applications of AI
- ▶ Introduction to key AI domains such as Data Science, Computer Vision, and Natural Language Processing
- ▶ Activities and data-based tasks
- ▶ Reflection on ethical use of AI

The AI content progresses from introduction to application, including introductory predictive techniques such as regression, classification, and clustering. The book emphasises on ethical and responsible use of AI, including introduction to bias, fairness, privacy, and safe use of technology, enabling informed and thoughtful engagement with AI systems.

Teachers should approach the book with the mindset that the process of thinking is more important than arriving at the correct answer. Creating a safe and encouraging environment where students feel comfortable making mistakes, exploring multiple strategies, and expressing their reasoning is essential. The goal is to nurture confident, independent thinkers rather than focus solely on correctness.

PART-1

COMPUTATIONAL THINKING

Chapter 1: Large Numbers Around Us

1. 'X' is a 6-digit number formed using exactly three different digits. One of the digits appears once, another appears twice, and the third appears three times. When 2 lakhs are added to 'X', the resulting number 'Y' is still a 6-digit number. What is the highest possible value of 'Y'?

a) 977889 b) 999999 c) 999988 d) 999887

2. What is the product of all the numbers on a telephone dial-pad?

a) 12345 b) 32451 c) 362880 d) None of these

3. Sam forms two different 6-digit numbers, X and Y, using digits from 0 to 9 (without repetition within a number). Exactly two digits are common between X and Y.

- The digit 9 appears in the same position in both numbers
- Y ends with 0, has only one even digit, and its digits are arranged in descending order
- 6 is present in X, and the number of even digits on its left is equal to the number of even digits on its right

If X is the largest possible number, what is the difference between X and Y?

a) 9310 b) 9210 c) 8420 d) 8310

4. If each of the rows follows the same theme, what will come in place of “?”

$$\boxed{\begin{array}{ccccccc} \textcircled{1} & \textcircled{} & \textcircled{8} & \textcircled{} & \textcircled{2} & \textcircled{} & \end{array}} > \boxed{\begin{array}{ccccccc} \textcircled{1} & \textcircled{} & \textcircled{} & \textcircled{8} & \textcircled{2} & \textcircled{} & \end{array}}$$

$$\textcircled{} \leq 7$$

$$\boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{} & \textcircled{3} & \textcircled{5} & \textcircled{} & \textcircled{7} & \end{array}} < \boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{3} & \textcircled{5} & \textcircled{} & \textcircled{} & \textcircled{} & \textcircled{7} & \end{array}}$$

$$\textcircled{} \leq 3$$

$$\boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{9} & \textcircled{6} & \textcircled{} & \textcircled{3} & \textcircled{} & \end{array}} > \boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{9} & \textcircled{6} & \textcircled{} & \textcircled{} & \textcircled{} & \textcircled{3} & \end{array}}$$

$$\textcircled{} \leq 2$$

$$\boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{} & \textcircled{5} & \textcircled{0} & \textcircled{6} & \textcircled{} & \end{array}} < \boxed{\begin{array}{ccccccc} \textcircled{} & \textcircled{5} & \textcircled{} & \textcircled{6} & \textcircled{0} & \textcircled{} & \end{array}}$$

?

$$\textcircled{} \leq 6$$

a)

$$\textcircled{} \leq 4$$

b)

$$\textcircled{} \leq 5$$

c)

$$\textcircled{} \leq 9$$

d)

5. What will come in place of “?”

Crore - re	→	00000
------------	---	-------

Million - ion	→	000
---------------	---	-----

Thousand - d	→	00
--------------	---	----

Billion - llion	→	0000
-----------------	---	------

Lakh - kh	→	?
-----------	---	---

a) 0

b) 00

c) 000

d) 0000

6. Six number tokens are given. Rearrange them to form a 6-digit number such that:
- The difference between the first and last digits is as small as possible
 - The hundreds digit is double the thousands digit
 - No consecutive digits appear next to each other
 - All given tokens must be used exactly once

LEFT **2** **5** **8** **1** **4** **7** RIGHT

- a) 1 b) 2 c) 3 d) More than 3

Number 1: 1 9

No two consecutive digits of the number series are present in the same number.

- a) 47525 b) 52663 c) 52429 d) 46733

- C has an even number of zeros in his number
- A's number has 2 zeros less than C's number
- D's number has 4 zeros more than A's number
- The number of zeros in B is more than that of A but less than C

- a) One thousand b) Ten thousand c) One lakh d) Ten lakh

- The cricketer earns the highest
- Alex earns more than Bob, while the doctor earns more than David, who is the engineer
- Bob is the lawyer, and neither Bob nor David has an exact salary of ₹50,000

- a) Cricketer - ₹70,000 b) Doctor - ₹60,000 c) Doctor - ₹50,000 d) Doctor - ₹40,000

35 tens

25 ones

14 ten thousands

- a) 213625 b) 203675 c) 212375 d) 213675



The Thinking Spot

Columns A, B, and C contain 3 numbers each, such that:

- The number of times each number appears in all 3 columns combined, is equal to its value (e.g., 2 appears twice, 3 appears thrice, etc.)

- Each column contains AT LEAST TWO different numbers

If column A is the only column with THREE different numbers, then which column has the highest sum?

A	B	C
○	2	○
○	○	○
4	○	3

(a) Column A

(c) Column C

(b) Column B

(d) Both columns B and C



Chapter 2: Arithmetic Expressions

1. Identify the odd one out from the following expressions, after INTERCHANGING their operators.

a) $204 + 3 \div 29$

b) $67 - 2 \times 37$

c) $48 \div 6 \times 3$

d) $195 \times 49 - 2$

2. There is a seminar from 11:00 AM to 2:00 PM, where students must spend AT LEAST 1 hour in the seminar. The table displays the number of students entering and exiting the seminar at different times. What is the MAXIMUM possible number of students who spent at least 2 hours in the seminar?

SR. NO	TIME	IN	OUT
1.	11:00 AM	120	
2.	12:00 PM	40	60
3.	1:00 PM	80	40
4.	2:00 PM		140

a) 60

b) 80

c) 100

d) 120

3. In the given expression, ' $8 @ 7 - 15 < 4 \times 2 \times 5$ ' which operator cannot be used in place of '@', such that the condition remains valid?

a) -

b) $\div$

c) +

d) $\times$

4. A bus has 40 seats, and some seats are already occupied, with one passenger in each seat.

- At the first stop, 5 passengers get off and 8 passengers get in
- At the second stop, 2 passengers get off and 4 passengers get in
- At the third stop, exactly half of the passengers get off

After this, half of the bus seats are occupied.

Which of the following expressions represents the number of passengers on the bus at the beginning?

a) $5 \times 5 + 5$

b) $6 \times 6 - 4$

c) $5 \times 5 + 10$

d) $6 \times 6 - 5$

5. The following expression has a whole number 'n'. If a blank can either take ' $\times 4$ ' or ' -12 ', what is the least possible value of 'n'?

Expression: $(n \text{ } \underline{\hspace{1cm}} \text{ } \underline{\hspace{1cm}}) \text{ } \underline{\hspace{1cm}} = 96$

a) 9

b) 18

c) 4

d) 72

6. Which operators should be interchanged to get the minimum value of the given expression?

$8 \times 4 + 2 - 1$

a) $\times$ and +

b) + and -

c) - and $\times$

d) No change

7. In certain code, 1 is coded as 2, 3 is coded as 10, 5 is coded as 26.

In the same way, 4 is coded as A and 6 is coded as B.

If C is the sum of A and B, which of the following is correct regarding C?

a) $C > 55$

b) $C < 50$

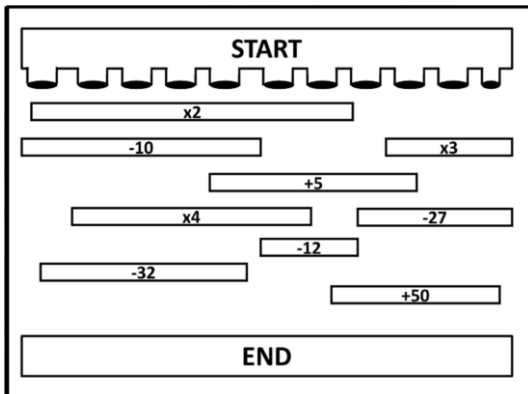
c) $C = 50$

d) $C < 55$

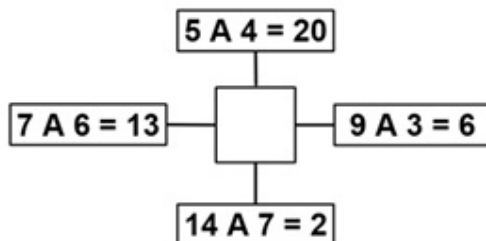
8. In the following expression, each ball with a mathematical operator is swapped with the ball immediately to its left, to form a new arithmetic expression. What will be the value of NEW EXPRESSION - ORIGINAL EXPRESSION?

LEFT $(8)(7)(+)(1)(5)(-)(3)$ RIGHT
Expression

- a) 63 b) 73 c) - 63 d) - 73
9. A stone with an initial value of 10 is dropped through one of the holes at the START and moves in a straight path to reach the END. As it passes each numbered bar along the way, its value changes according to the operation shown on that bar. What is the maximum value the stone can have when it reaches the END?



- a) 60 b) 88 c) 53 d) 68
10. Each arrow has a mathematical operator. When placed in the centre box, its operator replaces the letter A in the equation it points to. Which arrow satisfies an equation based on this rule?



- a) b) c) d)



The Thinking Spot

Tom, Sam, Mariya, John, and Bob each selected a shape. They placed their shapes in the grid below.

Tom's shape is neither a circle nor a triangle

Sam's shape is in column number 3

Mariya's shape is not in the first three rows

John's shape is neither square nor circle

Which of the following shapes is chosen by Bob?

	1	2	3	4	5
1	△				
2				○	
3		□			
4			☆		
5					⬡

(a) Square

(b) Circle

(c) Hexagon

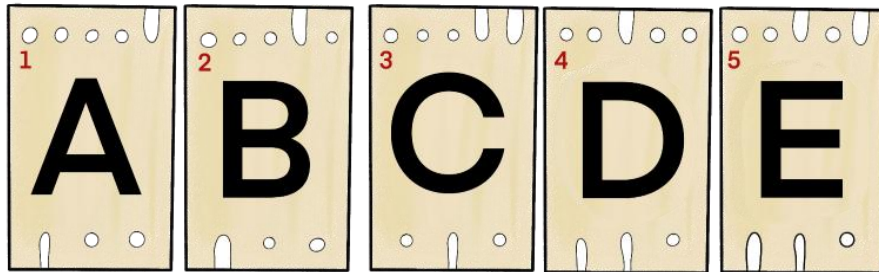
(d) Triangle



Chapter 3: A Peek Beyond the Point

Activity Time

Activity: Punched Cards



Consider a set of cards with the alphabet written on them, and we shuffle them. The objective is to arrange them in alphabetical order.

One method is to find the card A first, then B and so on. This might take some time.

Now, notice that the cards have been punched on the top. There are an O-shaped and a U-shaped punch, and using these holes, we can sort the cards in an interesting way. Insert a stick in the set of cards starting from the rightmost hole. Once you lift the stick, some will come along with it. Now, bring them in front & repeat this for all sets of holes. Magically, the cards are sorted.

In our day-to-day life, we use the decimal number system, which uses the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9. The decimal number system is based on bundles of ten (Powers of 10 which is 1,10,100,1000,...). Similarly, the binary number system is based on bundles of two (Power of 2 which is 1,2,4,8,...). So, if we compare the two numbering systems for place values, this is what it looks like...

	Decimal (Bundles of Ten)	Binary (Bundles of Two)
Units Place	1	1
First Bundle	$10 = 10$	$2 = 2$
Second Bundle	$10 \times 10 = 100$	$2 \times 2 = 4$
Third Bundle	$10 \times 10 \times 10 = 1000$	$2 \times 2 \times 2 = 8$

The binary number system (popularly known as the language of computers) deals with only two numbers: 0 and 1. We can convert any decimal number to its binary equivalent and vice versa. You can convert any decimal number into a binary representation by writing it with the power of 2.

For example, $23 = 16 + 4 + 2 + 1$.

23 in decimal	$10^1 = 10$	$10^0 = 1$
23	2	3

23 in binary	$2^4 = 16$	$2^3 = 8$	$2^2 = 4$	$2^1 = 2$	$2^0 = 1$
$(10111)_2$	1	0	1	1	1

1. What will be the binary representation for the decimal number 14?

- a) 1010 b) 1100 c) 1110 d) 1111

The number of digits or bits (binary digits) in a binary number depends on the number. For example, numbers up to 7 can be represented in a 3-bit binary number, numbers from 8 to 15 in a 4-bit binary number, numbers from 16 to 31 similarly need an additional bit, hence a 5-bit binary number, and so on. There are 26 letters in English; hence, 5 sets of holes on each card for sorting.

2. You want to perform a similar punched card activity with Hindi alphabets. How many holes are needed on each card?

- a) 5 b) 6 c) 8 d) 10

Punching Pattern on the Card

Every alphabet can correspond to a number, and every number can be represented in the binary system. The following table contains the corresponding conversion from decimal to binary.

Letter	Corresponding Decimal Number	Corresponding Binary Number
A	1	00001
B	2	00010
C	3	00011
D	4	00100
E	5	00101
..	..	..
..	..	..
Y	25	11001
Z	26	11010

These binary numbers indicate to us a unique way to punch. Where there is 0, punch an O-shaped hole, and where there is 1, punch a U-shaped hole. For example: 01010 is the binary equivalent of J, so the holes on card J from left to right will be O U O U O.

3. Can you identify which card the binary pattern 10101 belongs to?

- a) P b) S c) T d) U

4. The alphabet R comes 18th in the order starting from A as 1. What will be the punching pattern on Card R?

- a) U0U0U b) UU0U0 c) U00U0 d) 0UU00

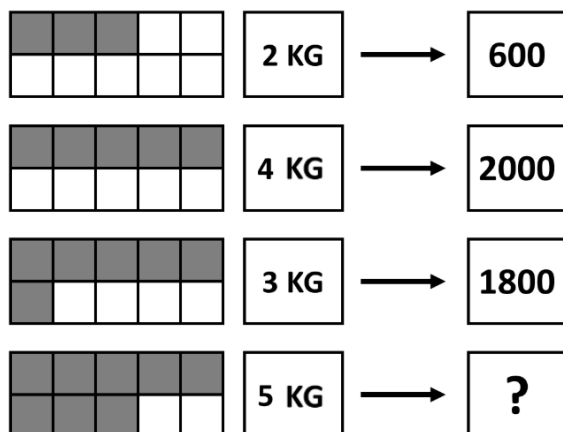
Punched cards, utilising binary choices is analogous to their role in powering early computing machines such as IBM sorters and Turing's Bombe. This binary concept has its roots in study of poetry in ancient India, which was well studied by Acharya Pingala's and documented in his work Chandaḥśāstra (~2 BCE).

Questions

1. Two decimal numbers differ by 3.6. In one of the numbers, the decimal point is moved one place to the right. After this change, both numbers become equal. Which was the smaller of the two original numbers before shifting the decimal point?

a) 0.4 b) 0.36 c) 3.6 d) 4

2. What will come in place of “?”



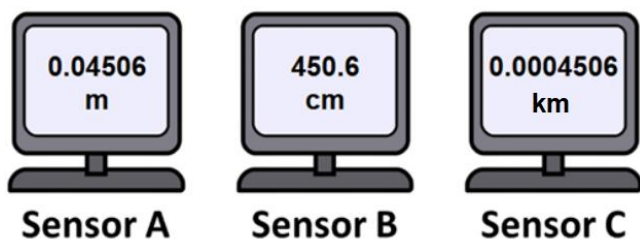
a) 2000 b) 4000 c) 40000 d) 8000

3. Find the next term in the given series below:

85.41, 88.71, 89.01, 92.31, 92.61, 95.91, 96.21, __

a) 96.51 b) 99.21 c) 99.51 d) None of these

4. There are three sensors - A, B, and C. Each sensor displays the measurement of the length of the same item, but in different units: metres, centimetres, or kilometres. Only one sensor shows the correct length. The other two show values that are 10 times greater and 10 times smaller than the actual length (the units remain unchanged). Based on the readings shown, which sensor displays the correct length?



a) Sensor A b) Sensor B
c) Sensor C d) Cannot be determined

5. What will come in place of “?” in the given series?

5	4
4.5	

,

4	4.5
4.25	

,

4.5	4.25
4.375	

,

4.25	4.375
4.3125	

,

4.375	4.3125
4.34375	

,

?	
---	--

a)

4.3125	4.450
4.38125	

b)

4.3125	4.34375
8.65625	

c)

4.3125	4.34375
4.328125	

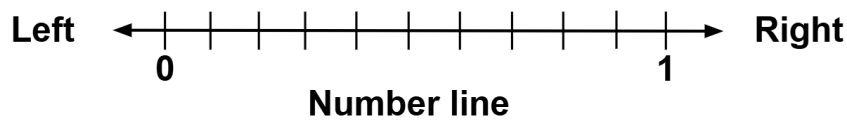
d)

4.3125	4.375
4.32812	

6. Four friends A, B, C, and D each select a distinct point on the following number line (from 0 to 1) such that:

- A's point is 4th to the right of D's point
- The value of C's point is at least 0.4 and B's point has the highest value among all friends
- No two friends choose adjacent points

What is the LEAST possible sum of the points chosen by all of them?



- a) 2.0 b) 1.9 c) 1.8 d) 1.4

7. Alex has to fill in the blank with a DECIMAL NUMBER formed using every digit among 0, 1, 2, 3, and 4, each exactly once, where:

- The sum of the digits before the decimal point is equal to the sum of the digits after the decimal point
- The digits before the decimal point are in descending order and the digits after the decimal point are in ascending order
- The conditions of “<” are satisfied

How many different numbers can be placed in the blank?

$$34.201 < \underline{\hspace{2cm}} < 412.03$$

- a) 2 b) 3 c) 4 d) More than 4

8. Raj measures the lengths of two objects, Object A and Object B, using a ruler marked in centimetres.

- On Day 1, Object A measures 5.2 cm. Its length decreases by 1 line on Day 2, 2 lines on Day 3, 3 lines on Day 4, and so on, following the same pattern
- On Day 1, Object B measures 1.2 cm. Its length increases by 1 line on Day 2, 2 lines on Day 3, 3 lines on Day 4, and so on, following the same pattern

On which day will the difference between the lengths of the two objects be the least?

Note: The scale is in centimetres



- a) Day 3 b) Day 4 c) Day 5 d) Day 6

9. Some numbers are given in the boxes below. In each number, interchange the digits immediately to the left and right of the decimal point. After making this change, how many boxes will contain a number greater than 498.540?

491.893	497.860	491.855	490.993
A	B	C	D

- a) 1 b) 2 c) 3 d) 4

10. Seven decimal numbers, A-G, are given.

1. Arrange the numbers in ascending order to form Arrangement 1
 2. Modify the numbers as follows:
 - For numbers in odd positions, move the decimal point one place to the right
 - For numbers in even positions, move the decimal point one place to the left
 3. Arrange the modified numbers again in ascending order to form Arrangement 2
- In Arrangement 2, which pair(s) of numbers have exactly three numbers between them?

0.05, 0.505, 5.05, 5.55, 0.55, 5.005, 0.005

A B C D E F G

- a) A and E b) G and D c) C and F d) E and D



The Thinking Spot

The numbers given below are to be placed in the grid in such a way that the sum of two adjacent numbers in the grid should be greater than 9 but less than 17. What will come in place of A and B?

Numbers: 2, 3, 5, 11, 15

14			A	8	B	12	1	
----	--	--	---	---	---	----	---	--

- (a) A = 5 and B = 2 (b) A = 3 and B = 5
(c) A = 2 and B = 3 (d) A = 5 and B = 3



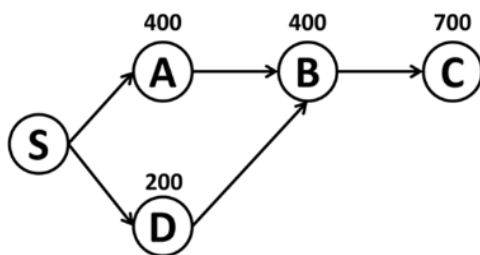
Chapter 4: Expressions using Letter-Numbers

1. It is given that x means 'greater than', $\$$ means 'equal to', $<$ means 'not less than', A means 'less than' and $+$ means 'not greater than'. Which of the following options has the same meaning as the given expression?

$a \times b A c$

- a) $a \$ c A b$ b) $b < a \times c$ c) $c \times b A a$ d) $c + b < a$

2. A factory 'S' sends goods to four cities: A, B, C, and D through pipelines (represented by arrows), as shown below. The demand of each city is mentioned above that city (for example, the demand at City A is 400 units). The demands are satisfied exactly, and no extra units are left in the pipelines. If 300 units are sent through the pipeline from A to B, how many units are sent from D to B?



- a) 200 b) 600 c) 800 d) 1000

3. Each letter in Set 1 corresponds to exactly one different number from Set 2 (in any order).

Set 1: A, B, C, P, Q, R

Set 2: 1, 2, 4, 5, 7, 9

The following conditions are given:

- A and C are consecutive numbers, with $A > C$
- $P + Q = R$, with $P > Q$
- B is greater than both A and C

Using this information, find the value of: $(4P+3B) - (2P+3C) + (2A-3B)$

- a) 9 b) 13 c) 15 d) 11

4. The sum of the ages of four people A, B, C, and D is 35. Each one of them is at least 8 years old. What is the **MAXIMUM** possible age of the oldest person?

- a) 9 years b) 10 years c) 11 years d) 12 years

5. 3 whole numbers A, B, and C, satisfy the given conditions. At maximum, how many different possible values can C have?

$$A > B > C \text{ and } A + B = 10$$

- a) 4 b) 5 c) 3 d) 2

6. If the sum of the ages of Shubham and Shivam is 36. What would be the sum of their ages after 36 years?

- a) 108 b) 136 c) 72 d) 96

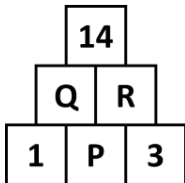
7. A, B, and Z are natural numbers such that $A + B = Z$ and:

- The value of B is twice the value of A
- The value of B is greater than 4
- The value of Z is less than 10

What could be the value of A?

- a) 1 b) 2 c) 3 d) 4

8. In the given image, each number equals the sum of the two numbers immediately below it. If P, Q, and R are unknown, what is the value of $P \times Q + R$?



- a) 42 b) 48 c) 38 d) 36

9. 5A, 5B, and C9 are 2-digit numbers, where A, B, and C represent one of the digits of the respective numbers, and $A + B + C = 15$. Find the MAXIMUM possible value of $3B + 4C + 8$, if:

$$5A > 5B > C9$$

- a) 43 b) 39 c) 31 d) 36

10. All the cells having 'x' in the given calendar represent dates that are consecutive multiples of a particular number. The first 'x' represents the first Thursday of the month, as shown below. Which of the following is the last day of the month?

November						
MON	TUE	WED	THU	FRI	SAT	SUN
			x			
		x				
	x					
x						

- a) Monday b) Sunday c) Saturday d) Friday



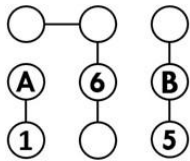
The Thinking Spot

Numbers from 1-9 are to be placed in the grid such that no two consecutive numbers are placed adjacent to each other.

For example, the bottom-left circle has 1, so the circle above it and the circle on the immediate right cannot have the number 2.

Each group of circles which are inter-connected by lines must contain either only odd or only even numbers.

Find the values of A and B.



- (a) A = 9 and B = 3 (b) A = 3 and B = 9 (c) A = 7 and B = 3 (d) A = 9 and B = 8

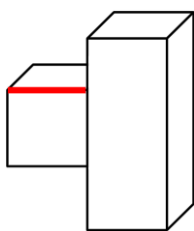


Chapter 5: Parallel and Intersecting Lines

1. Each side of a square is divided into three equal parts by marking points on it. Using these points as endpoints, line segments are drawn inside the square, such that each line segment is parallel to at least one side of the square. What is the maximum number of squares that can be found in the resulting figure?

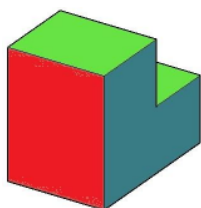
a) 6 b) 9 c) 14 d) 16

2. The following arrangement is formed using a cube and a cuboid. How many edges are parallel to the edge highlighted by the red line?



a) 5 b) 6 c) 7 d) 8

3. In the following solid, every edge has at least one other edge parallel to it. How many different pairs of PARALLEL FACES are present in the solid?

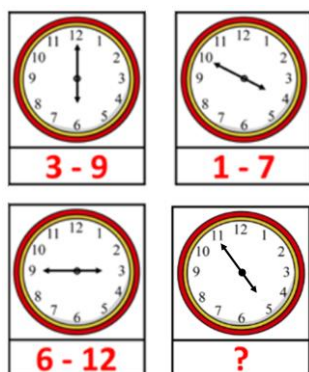


a) 8 b) 7 c) 6 d) 5

4. In a clock, the hour hand and the minute hand are perpendicular to each other, where the minute hand is at 12. How many different possible positions the hands of the clock can have at this time?

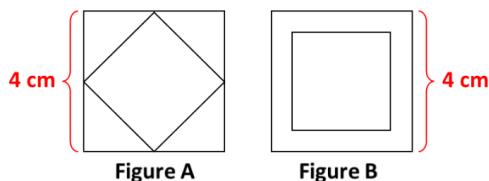
a) 1 b) 2 c) 3 d) 4

5. If each of the given terms follows the same theme, what will come in place of “?”

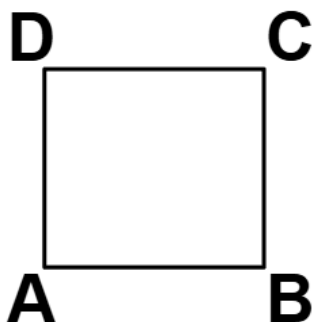


a) 7 - 1 b) 2 - 8 c) 1 - 8 d) 5 - 11

6. Figures A and B are formed using two transparent squares, each. If figure A is placed on top of figure B (without rotation), such that it completely overlaps figure B, how many right-angled triangles can be seen in the resultant image?



- a) 10 b) 12 c) 13 d) 14
-
7. In triangle ABC, BC is the base.
 Point P lies on AB and point Q lies on BC such that PQ is parallel to AC.
 Point R lies on AB and point S lies on AC such that RS is parallel to BC.
 Lines PQ and RS intersect at a point O inside the triangle.
 Which of the following quadrilaterals must be a parallelogram?
- a) QPAC b) ORBQ c) QOSC d) SAPO
-
8. Two lines intersect at point K.
 One angle is $(ax - 10)^\circ$ and its vertically opposite angle is $(bx + 20)^\circ$.
 Which of the following is DEFINITELY true?
- a) $a = b$ b) $a \neq b$ c) $x = 10$ d) $a + b = 0$
-
9. A computer draws two intersecting lines. One angle formed at the intersection is 70° , but originally, it was intended to be 40° . Which of the following options shows the respective 'Expected and Resultant' measures of one of the other angles?
- a) 140 degrees and 110 degrees b) 110 degrees and 140 degrees
 c) 140 degrees and 170 degrees d) 170 degrees and 140 degrees
-
10. ABCD is a square of side length 4 m. Raj starts from point A and walks North to reach point D. From point D, he further walks 4 m without taking any turns and reaches point P. At P, he turns 135° clockwise and walks a certain distance and stops only when he reaches a point on the square. Which of the following statements is INCORRECT with respect to the boundary of the square and the path taken?



- a) $PC \parallel DB$ b) $\angle PCD = \angle DCA$ c) $PC \neq AC$ d) $\angle DPC = \angle DCP$



The Thinking Spot

A, B, and C are all different numbers having any value among 1, 2, and 5. What will come in place of C so that no two consecutive numbers are placed next to each other in any direction (vertically, horizontally, or diagonally)?

	C	3	
B	8	A	7
	6	4	

(a) 2

(b) 1

(c) 5

(d) None



Chapter 6: Number Play

Activity Time

Introduction

Solving puzzles always seems complex. But have you wondered how computers solve puzzles? We will see a simplified way to solve some puzzles by following simple instructions and using the binary computer language. This will give us insight into how we can break complex problems into smaller pieces that can be easily solved. By combining smaller problems, computers can solve a larger one.

Activity: Logical puzzle punch cards

In this activity, you will use punched cards to solve logical puzzles. By stacking, aligning, and filtering the cards, we will see how complex reasoning can be built from simple decisions - **True** or **False**, mirroring how binary logic works inside a computer.

The scenario is that you have the following lights in the room: **Yellow Y**, **Pink P**, **Red R**, **Green G**, and **Blue B**. By following the clues given in the **Case Cards**, you have to figure out which lights are **ON** in the given Puzzle. To solve these puzzles, we have a deck of cards representing all possible combinations of these lights being **ON** or **OFF**.

Understanding the cards

To solve these puzzles, we have a deck of cards representing all possible combinations of these lights being **ON** or **OFF**. Each card has five positions for holes on the top, and each position corresponds to a coloured light mentioned above in the given order.

Each position has two possible hole types: **U-shaped** or **O-shaped**. Each card represents one possible combination. If a light is **ON** in a card, then the hole corresponding to it is **U-shaped**. If a light is **OFF** in a card, then the hole corresponding to it is **O-shaped**.

Eg: **Yellow Y** is **OFF**, **Pink P** is **ON**, **Red R** is **ON**, **Green G** is **OFF**, **Blue B** is **OFF** is the card where the holes are **O U U O O**.

Each hole, thus, has 2 possible states. If we have two holes, then we have $2 \times 2 = 4$ possible states. Likewise, if we have 3 holes, we have $2 \times 2 \times 2 = 8$ states. Therefore, if we have 5 holes, we will have $2 \times 2 \times 2 \times 2 \times 2 = 32$ states. That is the number of distinct cards we have.

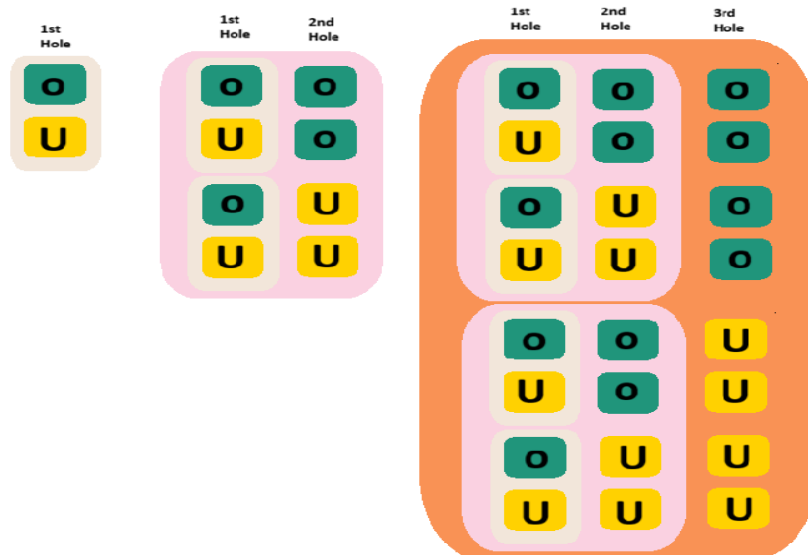


Figure: All the possible three-hole states and how they come from the two-hole states

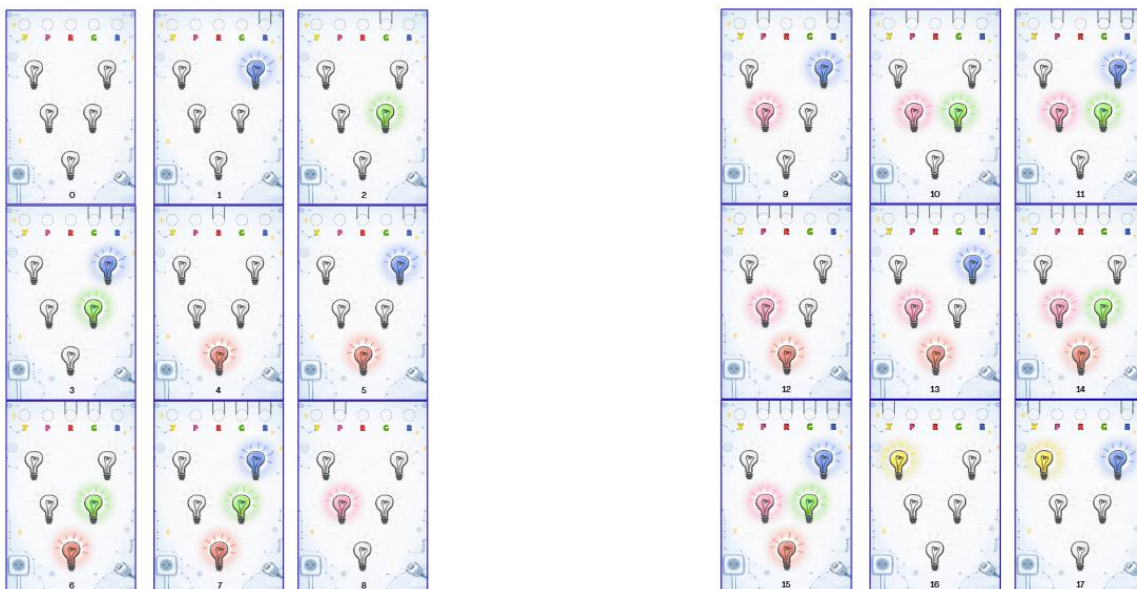


Figure: Some possible cards with 5 binary digits

Procedure

If the puzzle says a light is **ON**, we need to remove all cards where that light is **OFF**. For that, we put the stick into the deck of cards through the hole corresponding to that light and pull out all the cards that get stuck to it. We remove these cards because they got stuck due to **O-shaped** holes that indicate the light is **OFF**.

If we know some light is **OFF** in the puzzle, we remove all the cards that do not come out with the stick; these are the cards with **U-shaped** holes in positions corresponding to the given colour, which indicate that the light is **ON**.

Ensuring the above helps us solve the logical puzzles through punch cards.

Let's get familiar with the cards and the rules.

1. Can you identify which bulbs are 'OFF' based on the card?



- a) **Blue B** and **Yellow Y** bulbs
b) **Green G** and **Red R** bulbs
c) **Red R** and **Yellow Y** bulbs
d) All bulbs are OFF

Understanding using a case

The operation of the cards is explained by showing how they can be used for solving a problem in two-valued logic.

We can start by taking **Case 1**, which contains the first 8 cards from 0 to 7, and focus only on the **Red R**, **Green G**, and **Blue B** lights. Three statements are given containing simple assertions and negation statements. The aim is to determine which bulbs are on or off among these.



To solve this problem, let's establish some notation. Only three terms are involved, so we shall be concerned with only the **Red R**, **Green G**, and **Blue B** holes.

Red bulb ' ON ', 1st position will be U	Red bulb ' OFF ', 1st position will be O
Green bulb ' ON ', 2nd position will be U	Green bulb ' OFF ', 2nd position will be O
Blue bulb ' ON ', 3rd position will be U	Blue bulb ' OFF ', 3rd position will be O

If all lights are **ON**, then **RGB = U U U**

If all lights are **OFF**, then **RGB = O O O**

If **Red** is **OFF**, **Green** is **ON**, **Blue** is **OFF**, then **RGB = O U O**

If **Red** and **Green** are **ON**, but **Blue** can be anything, then **RGB = U U X**

2. Can you identify which bulb is OFF based on this pattern: **RGB = U U O**?

- a) **Red** bulb
b) **Green** Bulb
c) **Blue** Bulb
d) All bulbs are OFF

Case 1: Focusing on RGB cards only (with 8 cards only)

Statements

I. The **Red** light is **ON**

II. The **Blue** light is **OFF**

III. **ON** light only has **OFF** Neighbour

The problem has three statements.

- *The statement I: "The **Red** light is **ON**"*

Insert the stick in the **R** position and lift.

The stick will raise all **RGB = O X X** cards: All cards where **Red** is **OFF**.

RGB = O X X cards are discarded.

All remaining cards are assembled into a pack once more (in any order), and we are ready for the second statement.



3. After performing an action using Statement I, how many cards are left in the pack?

- a) 8 b) 4 c) 2 d) 1

- *The second statement, II, "The **Blue** light is **OFF**"*

4. After performing the action using Statement II, among these, which card will not be discarded?

- a) **RGB = X X O** b) **RGB = U O U** c) **RGB = X X U** d) All will be eliminated

Correct answer: a. **RGB = X X O**

Explanation: We discard all the cards where **Blue** light is **ON**, **RGB = X X U**, and keep those cards where **RGB = X X O**.

Competencies: Algorithmic thinking, decomposition, pattern recognition

All remaining cards are assembled into a pack once more (in any order), and we are ready for the third statement.

- *The third statement, III, "ON light only has OFF Neighbour"*



After this action, only one card will be left, and that will be the solution to the logical puzzle.

5. What is the pattern of the last card (solution)?

a) RGB = U U O

b) RGB = U O U

c) RGB = O U U

d) RGB = U O O

Case 2: Focusing on RGB cards only (with 8 cards only)

Statements

I. The **Red** light is **ON**

II. The Lights are either all **ON** or all **OFF**



6. Among these, what would be the pattern of the solution?

a) RGB = U O O

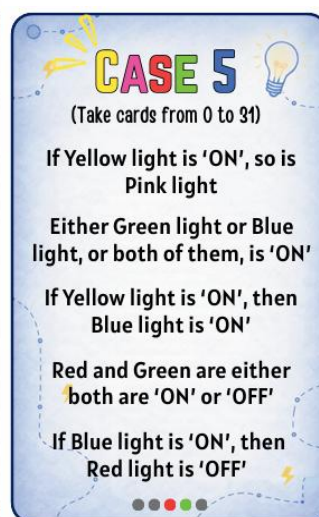
b) RGB = O O O

c) RGB = U O U

d) RGB = U U U

More Puzzles

Below are three progressively harder cases. They introduce additional logic and multi-step filtering to mimic advanced truth-table-based reasoning. Each ends with one solution card.



Questions

1. Form the largest four-digit odd number by selecting exactly one digit from each row and each column. What is the sum of the digits?

7	4	5	3
6	9	2	6
5	1	3	2
8	9	7	4

- a) 21 b) 22 c) 23 d) 24

2. In the given image, S, T, and N represent distinct single-digit numbers, where $T > S$. What is the value of $N + S$?

	S	T	6
+	T	S	N
1	7	6	8

- a) 13 b) 12 c) 9 d) 11

3. Each square of the given puzzle must be filled with a single-digit natural number such that:
- The sum of all four different digits in each row and each column is 10
 - All the inequality signs ($<$ and $>$) between the boxes must be followed
- Find the correct values of A and B.

1			4
^		^	
2	B		A
	3		^
	^	^	^
		2	

- a) $A = 3, B = 4$ b) $A = 2, B = 1$ c) $A = 1, B = 4$ d) $A = 3, B = 1$

4. How many cells contain a digit equal to the sum of all the other digits in its corresponding row and column combined?

3			6
1	2	3	
	4		2
5			1

- a) 2 b) 3 c) 4 d) 5

5. Using the digits 1, 2, 3, and 4, form the highest and the smallest three-digit numbers such that each number contains exactly two different digits. What is the sum of the two numbers formed?

- a) 545 b) 555 c) 565 d) 575

6. A 2-digit number is formed using two different digits, and the digits are reversed to create another 2-digit number. The difference between these 2 numbers CANNOT be:

- a) An even number b) An odd number c) A prime number d) An integer

7. A four-digit number has the same number of even and odd digits. Which of the following could possibly be the sum of the digits of this number?

- a) 11 b) 15 c) 20 d) All of these

8. The given grid is to be filled with distinct single-digit natural numbers such that:

- The sum of each row is the same and is an even number
- The larger number in each column is placed in the red block

What is the sum of the numbers in the white blocks?

		7
4	8	

- a) 16 b) 17 c) 15 d) 20

9. A computer generates rectangles where the length and breadth (dimensions) follow a series. If the pattern continues, what will be the area of the next rectangle the computer should draw?

$2 \times 3, 3 \times 5, 5 \times 8, 8 \times 13, 13 \times 21, ?$

- a) 693 sq. units b) 2210 sq. units c) 442 sq. units d) 714 sq. units

10. Given below is a 3×3 grid to be filled by numbers from 1 to 9 such that:

- Every pair of adjacent digits in a row should have a difference of three
- Some of the numbers have already been placed

Based on the given conditions, choose the correct option.

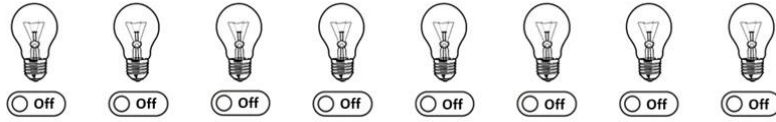
1		
	5	
		9

- a) A row can have all numbers as even b) A row can have all numbers as odd
c) A row can have 2 even and 1 odd number d) All of these



The Thinking Spot

Given below are bulbs, each with a switch beneath it. When you press a switch, it lights up **ONLY** the adjacent bulbs and not the bulb above the switch. What is the minimum number of switches you need to press to light up all the bulbs?



(a) 2

(b) 3

(c) 4

(d) 5

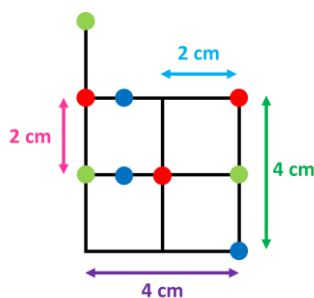


Chapter 7: A Tale of Three Intersecting Lines

1. If the sides of an equilateral triangle and that of a square are of equal length, what can you say about the area of both the shapes?
- The area of the square is the same as that of the triangle
 - The area of the triangle is greater than that of the square
 - The area of the square is greater than that of the triangle
 - None of these

2. Three identical pieces of paper, each in the shape of an equilateral triangle, are joined together edge to edge, with no gaps or overlaps. Which of the following figures can be formed by joining them?
- Rhombus
 - Trapezium
 - Hexagon
 - Triangle

3. If dots of the same colour are connected to form triangles, which triangle is not formed by any coloured dots?



- Right-angled triangle
- Obtuse angled triangle
- Equilateral triangle
- Isosceles triangle

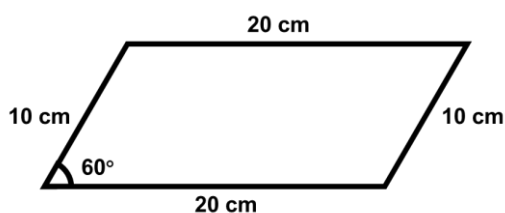
4. Is the given triangle isosceles, scalene, or equilateral?

Information 1: The given triangle is similar to a right-angled triangle

Information 2: The given triangle is not a scalene triangle

To answer the given question, which of the given information is/are sufficient?

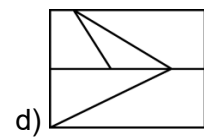
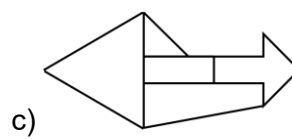
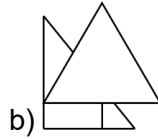
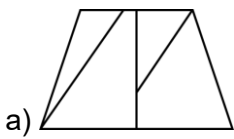
- Only 1
 - Only 2
 - Both 1 and 2 together
 - Question cannot be answered even if both pieces of information are used
5. At least how many folds are required to convert the given paper sheet into an equilateral triangle?



- 1
- 2
- 3
- 4

6. Each option contains exactly two types of triangles from the following: a right-angled triangle, an equilateral triangle, and a scalene triangle. Which option contains both a right-angled triangle and an equilateral triangle?

Note: Consider only those triangles whose all three sides are clearly visible



7. An isosceles triangle has a perimeter of 23 cm and all its side lengths are whole numbers. The two equal sides are multiples of 3. How many different possible lengths can the third side have?

- a) 1 b) 2 c) 3 d) 4

8. Two triangles, triangle X and triangle Y, are given. Only partial information about their corresponding sides and angles is provided below.

Question: Are triangles X and Y similar?

Information 1: Two pairs of corresponding sides of triangle X and triangle Y are proportional, but no information is given about the third pair of corresponding sides.

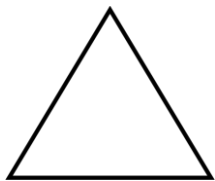
Information 2: Two pairs of corresponding angles of triangle X and triangle Y are equal, but no information is given about the third pair of corresponding angles.

To answer the given question, which of the given information is/are sufficient?

- a) Only 1
b) Only 2
c) Both 1 and 2 together
d) Question cannot be answered even if both pieces of information are used

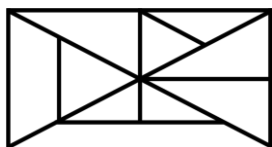
9. An equilateral triangle is given. Inside it, two straight lines are drawn in any direction. If the triangle is cut along these two lines, what is the maximum number of right-angled triangles that can be formed?

Note: You cannot form a triangle by joining two or more pieces.



- a) 2 b) 3 c) 4 d) 5

10. How many right-angled triangles are there in the given figure?

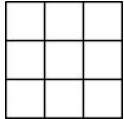


- a) 7 b) 8 c) 9 d) 10



The Thinking Spot

A 3 x 3 grid is shown below. You need to place some circles in the cells without making three in a row, column, and diagonal. How many circles at maximum can you place in the grid?



(a) 5

(b) 4

(c) 6

(d) 7



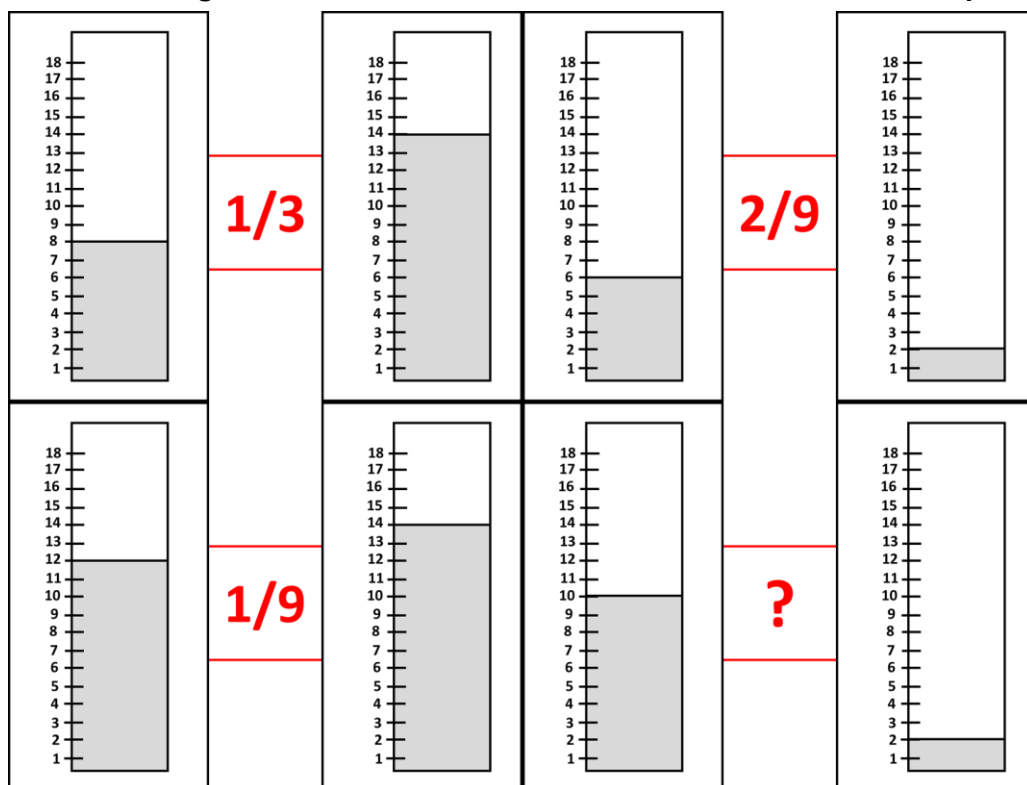
Chapter 8: Working with Fractions

1. Samuel received chocolates for his birthday. He decided to share it with his family.
- He keeps $\frac{1}{3}$ rd of the chocolates for himself
 - He gives $\frac{1}{4}$ th of what is left to his sister Sarah
 - He gives the remaining 12 chocolates (except the ones which he kept for himself) to his brother

How many chocolates had Samuel received in total before he shared it?

- a) 20 b) 22 c) 24 d) None of these

2. If each of the given terms follows the same theme, what will come in place of “?”



- a) $\frac{2}{9}$ b) $\frac{2}{3}$ c) $\frac{4}{9}$ d) $\frac{1}{9}$

3. A bus started its journey from Mumbai.
- $\frac{1}{4}$ th of the passengers got off the bus in Pune and 35 new passengers got on the bus
 - Half of the passengers got off the bus in Goa with no new passengers getting on the bus
- When the bus left from Goa, the total number of passengers was 85. How many passengers boarded the bus from Mumbai?

- a) 135 b) 170
c) 180 d) Cannot be Determined

4. Which of the following statements is/are sufficient to answer the given question?

Question: What fraction of the initial liquid remains in the vessel after Day 2?

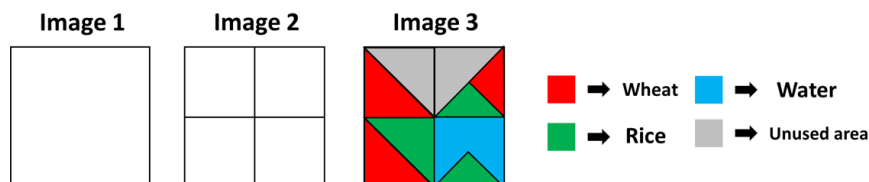
Statement 1: The vessel originally contained 100 litres of liquid

Statement 2: On Day 1, the amount of liquid that evaporates is equal to half of the quantity that would remain if one-third of the initial liquid were removed

Statement 3: On Day 2, the liquid left in the vessel becomes one-fourth of the quantity that was present at the beginning of Day 2

- a) Statement 3 alone is sufficient
- b) Both Statement 1 and Statement 2 are necessarily required
- c) Both Statement 2 and Statement 3 are necessarily required
- d) Question cannot be answered even if all the Statements are used

5. A farmer has a square farm as shown in Image 1. He divides it into 4 equal parts as shown in Image 2. Now, he allotted certain portions of land for specific purposes (Rice crop, Wheat crop, Water) and some part of the farm is unused, as shown in Image 3. What fraction of the new farm is occupied by wheat?



- a) $\frac{3}{16}$
- b) $\frac{1}{4}$
- c) $\frac{5}{16}$
- d) $\frac{6}{16}$

6. A wire is bent to form a rectangle with length L and breadth B. After some time, the length is increased by one-fifth of its original value, and the breadth is decreased by one-fifth of its original value. After this change:

- The perimeter of the rectangle remains the same
- The area of the rectangle decreases by 100 sq. cm

What was the original length of the rectangle?

- a) 25 cm
- b) 40 cm
- c) 45 cm
- d) 50 cm

7. A number sequence is mentioned below:

$\frac{19}{2}, \frac{38}{6}, \frac{76}{18}, \frac{152}{54}, \dots$

Which will be the first term of the series that will be less than 1?

- a) 7th term of the series
- b) 8th term of the series
- c) 10th term of the series
- d) None of these

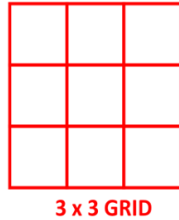
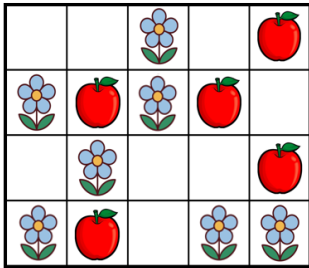
8. A box has 98 chalks. Each day, a teacher picks up a chalk to write.

When a chalk becomes $\frac{1}{7}$ of its original size, she sets it aside.

Whenever she collects enough such small pieces to make one full chalk, she joins them and uses it. For how many days will the 98 chalks last?

- a) 84
- b) 98
- c) 112
- d) 114

9. A grid contains apples and flowers in some of its cells. A 3×3 square frame can be placed on the grid without rotating and without extending outside the grid, covering exactly 9 cells. In how many different positions can the frame be placed so that at least one-third of the covered cells contain flowers?



- a) 3 b) 5 c) 4 d) 6

10. Each circle is divided into equal parts, with some parts shaded. A fraction is written to the right of each circle. How many of these fractions correctly represent the shaded part of their corresponding circles?



- a) 4 b) 3 c) 2 d) 5



The Thinking Spot

You have a collection of 10 different squares arranged as shown below. What is the minimum number of pairs of squares you need to interchange to ensure that white and black squares are arranged alternately?

Note: If you swap 1 black square with 1 white square, it is counted as 1 PAIR OF SQUARES being swapped



- (a) 1 (b) 2 (c) 3 (d) 5

